Bridging the Gap:

Comparing Employer and Educator Expectations in Small Animal Dentistry

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1 Abstract

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2 Introduction

Dr. Mariea Ross-Estrada, a faculty member at North Carolina State University's College of Veterinary Medicine, is exploring whether there are differences between the expectations of small animal primary care veterinary employers and veterinary educators regarding new graduates' competencies in dentistry. Through her own professional experience and conversations with colleagues, Dr. Ross-Estrada observed that many veterinarians must rely on on-the-job training to gain the skills necessary in small animal dentistry. These shared experiences prompted her to investigate whether there is a misalignment in what is taught in veterinary programs and what is expected in clinical practice.

To explore this question, Dr. Ross-Estrada distributed two surveys: one to medical directors and private practice owners and the other to primary care veterinary educators. Both surveys included similar questions regarding what early-career veterinarians are expected to have learned during their education and the skills they are expected to perform in practice.

This research has the potential to improve veterinary care by ensuring new graduates are better prepared to meet clinical demands. It could also support a smoother transition into practice for early-career veterinarians, reduce the burden of on-the-job training for employers, and promote overall workforce readiness in the field of small animal primary care.

2.1 Research Question

How do small animal primary care employers (medical directors and practice owners) and primary care veterinary educators differ in regards to their expectations of early career veterinary graduates' competencies in small animal dentistry?

2.2 Statistical Questions

1. Are there significant differences between educators and practice owners in their belief that new graduates are competent in key dental skills on their first day of practice?
2. Is there a difference between educators and practice owners in their reports (educators' actual teaching vs. owners' perceptions) of which dental skills were taught in the pre-clinical DVM curriculum for recent graduates?
3. Is there a difference between educators and practice owners in their level of agreement about whether specific dental skills should be taught pre-clinically?
4. Do employers and educators differ in their expectations about how many dental procedures new graduates should complete during clinical training?
5. Is there difference between the instructional formats in dentistry reported by DVM programs and the formats perceived by employers to have been completed by early career veterinarians?

6. Do educators and employers differ in their views on which formats of clinical instruction in dentistry should be required for DVM students as part of their clinical training?
7. Is there a difference between the clinical dentistry skills that educators report DVM students are learning during their clinical training and the skills that employers believe recent graduates have completed as part of their DVM program?
8. Do educators and employers differ in their opinions about which clinical dentistry skills DVM students should be required to practice or learn during their clinical training?

3 Data

3.1 Data Description

Two separate surveys were administered to mutually exclusive groups: veterinary employers who have worked with students, and educators who have taught students. There was no overlap between these groups and they can be assumed to be independent.

The employer data set consists of responses from 29 participants answering 40 questions, while the educator data set includes 43 participants answering 34 questions. Each group was asked a single qualifying question to determine eligibility for participation, along with nine questions covering demographics and institutional context. Educators were then presented with 24 competency and sentiment-based questions, while employers answered 30 such items focused on professional expectations and training in veterinary medicine.

Survey questions took several forms. Some were binary (Yes/No), particularly those related to demographics and institutional affiliation. Others used a “select all that apply” format, commonly seen in questions asking respondents to identify procedures performed at their practice. Many of these questions were followed by Likert-scale items. The Likert scales were even-numbered and omitted a neutral option, which may have contributed to at least two instances where respondents selected both “agree” and “disagree” for the same item.

Several questions offered an “Other” response with a text box for elaboration. A few required numeric input, such as estimates of hours worked or the number of practicing veterinarians. These integer fields were not restricted by any upper bound, regardless of contextual reasonableness.

Global survey session metrics

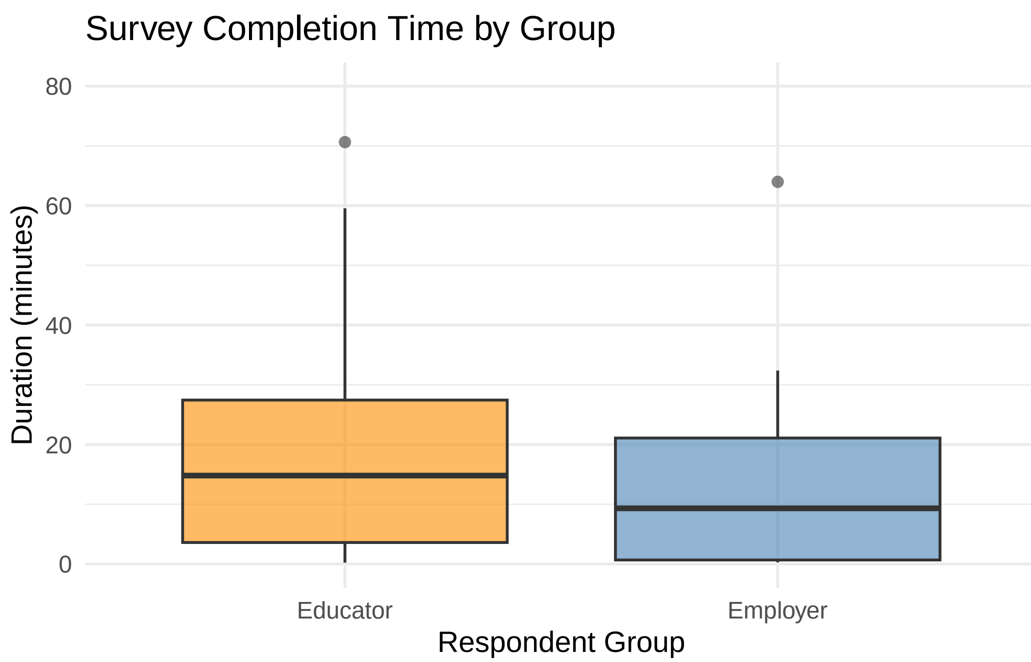

Figure 3.1: Assessing survey elapsed time distribution via box plots to understand engagement by survey group.

Survey completion time differed by group. Educators, on average, spent more time completing the survey than employers. While no follow-up question asked participants to explain their response time, this discrepancy may reflect greater engagement or a tendency for more elaborated responses among educators. It may also suggest a greater willingness among educators to participate more thoughtfully. The box plot below illustrates the distribution of survey duration (in minutes) by group.

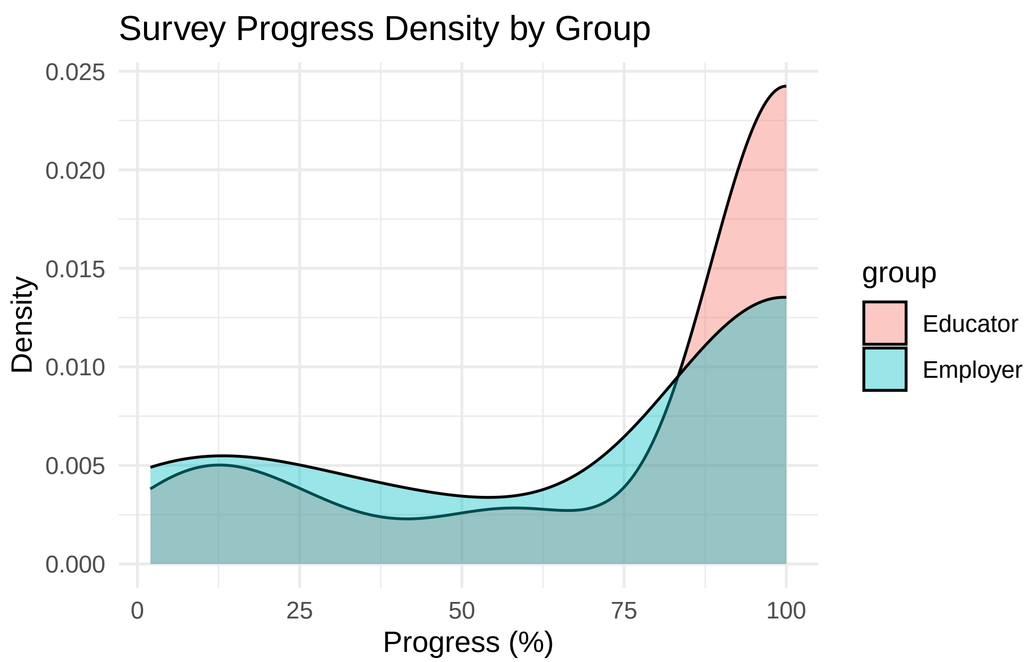

Figure 3.2: Assessing survey completion as a density curve to understand engagement by survey group.

Regarding the proportion of the survey completed, employer responses were more variable—spanning the full range from partial to full completion. In contrast, educators tended to complete more of the survey, with a concentration near full completion and a less pronounced left tail. The density plot below visualizes these differences in survey progress across groups.

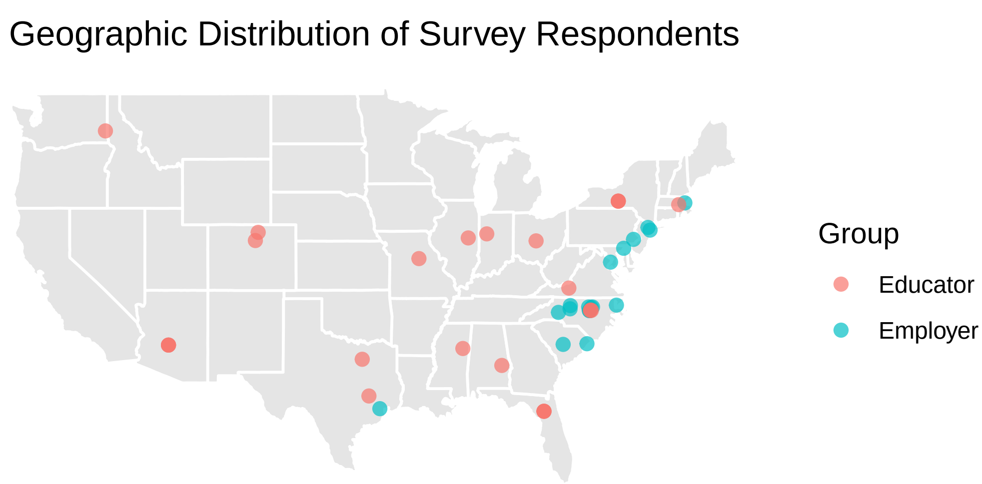

Figure 3.3: Geographic distribution of survey respondents across the United States.

Since our analysis focused on the representativeness of the United States, we identified three sets of coordinates in the educator dataset corresponding to international institutions: the University of Guelph in Ontario, Canada; Zanzibar University in Chake, Tanzania; and Chiba University in Chiba, Japan. Employer survey participants were predominantly sampled from locations along the eastern seaboard of the United States, while educators were more widely distributed across the country. We highlight this to illustrate that perceptions of veterinary dental students—particularly among employers—may differ for individuals located far from the regions where most participants were sampled. Additionally, because many of the responses came from major metropolitan areas, our findings may underrepresent opinions regarding student knowledge gaps in more rural settings.

Educator metrics

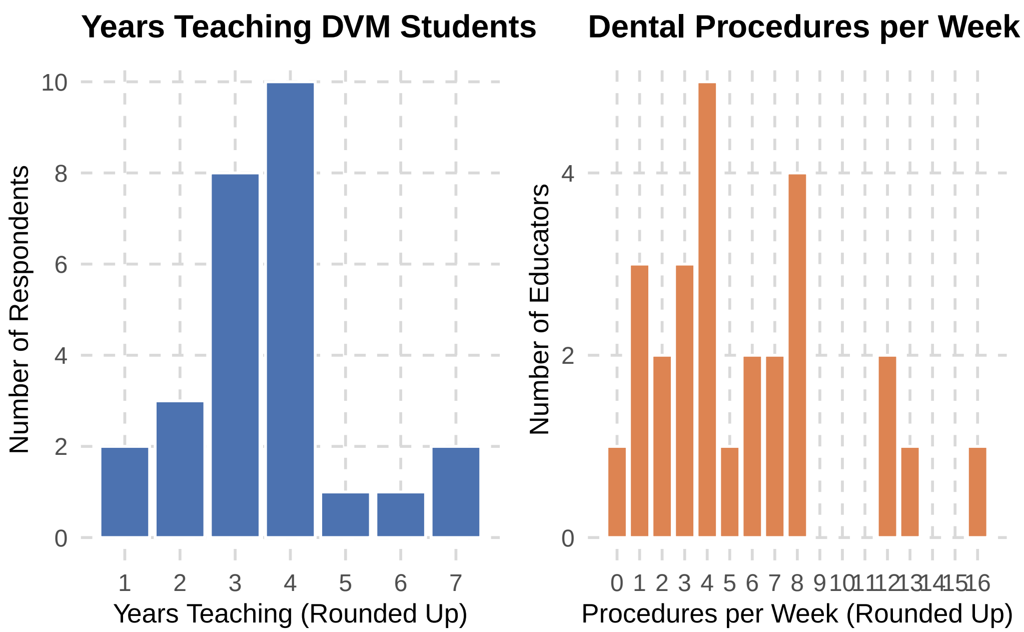

Figure 3.4: Educator contextualized background information about teaching and procedures.

Figure 3.3 provides an overview of the survey participants and their institutions. Most respondents reported having between 2 and 4 years of experience teaching veterinary students in clinical training. The distribution of years taught appears approximately normal, with fewer educators at the lower and upper ends of the experience range. Respondents also reported the number of dental procedures performed by their primary care service each week. While some outliers from busier institutions reported higher volumes, most educators estimated performing between 1 and 8 procedures per week.

Employer metrics

The employer dataset was considerably smaller than the educator dataset, which may reflect differences in motivation or willingness to complete the survey. Most employer respondents identified as practice owners, and approximately 75% indicated they were employed in private practice. This aligns with findings from the American Veterinary Medical Association (AVMA), which reported that "most veterinarians age 75 or younger, 83.9%, work in private clinical practice" (American Veterinary Medical Association 2019).

Table 3.1: Job Setting or Organization: Counts and Percentages

Job Setting	Count	Percentage
Group corporate veterinary practice	2	16.7
Independently owned group veterinary practice	2	16.7
Independently owned single veterinary practice	7	58.3
Industry/commercial	1	8.3

Table 3.2: Respondent Role: Counts and Percentages

Respondent Role	Count	Percentage
Associate veterinarian	2	18.2
Practice manager/HR representative	2	18.2
Practice owner	7	63.6

Survey responses were also heavily skewed toward practice ownership, with 63.6% of employer participants identifying as owners. By comparison, the same AVMA survey reported that only 21.3% of veterinarians identified as practice owners. This discrepancy may be due to the smaller sample size or the possibility that practice owners were more willing to complete the survey than associates or other veterinary professionals.

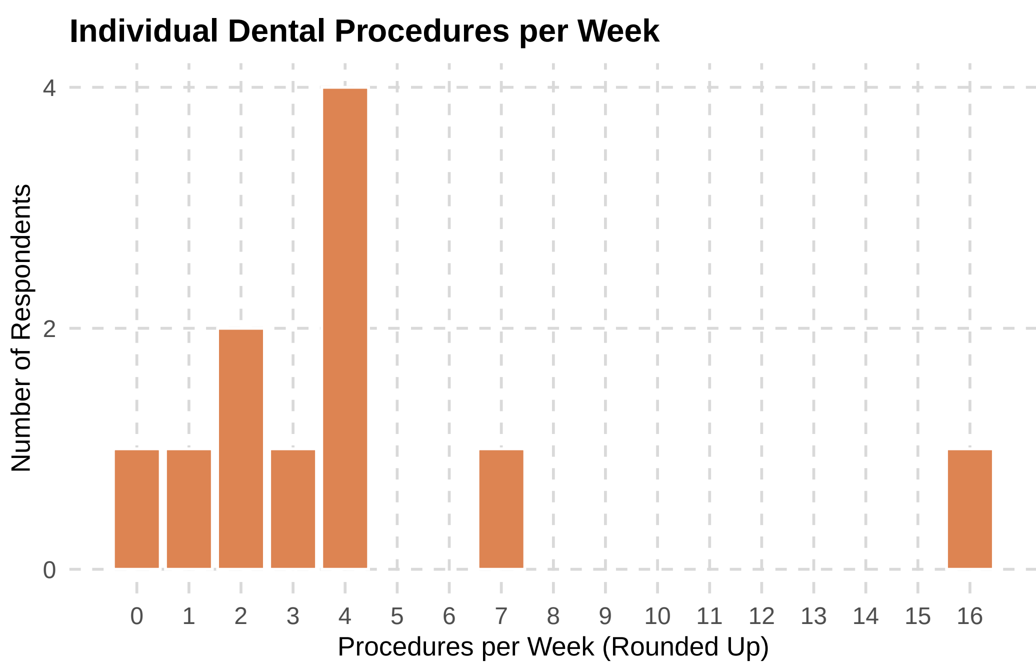

Figure 3.5: Average dental procedures employer survey participants indicated they performed each week.

Figure 3.5 shows that procedure counts tend to be lower in the employer group compared to the educator group, although the overall distribution shapes are similar. In both groups, the data exhibit a right-skewed pattern, with most respondents reporting lower procedure counts and a few outliers representing higher volumes. This pattern aligns with the intuitive understanding that while some practices or institutions have greater clinical demands on veterinarians, these cases are less common—at least based on the survey responses.

3.2 Data Source

Survey data were collected using Qualtrics, a cloud-based experience management platform commonly used for gathering feedback and sentiment across workforce domains. Participants from the educator survey were recruited via email invitation sent by the researcher, using pre-existing contact lists. Dr. Ross-Estrada distributed the employer survey to her personal and professional networks online. Participation was voluntary and anonymous. There was no incentive offered for completing the survey.

3.3 Preprocessing Description

Although the employer and educator data sets shared a similar structure, they were not identical. Most pre-processing steps were applied uniformly across both data sets, with minor deviations where needed.

The data sets were imported into the RStudio environment (version 2024.04.1 Build 748). A new variable was created to label the data source ("Educator" or "Employer") for later grouping and visualization. The existing respondent_id column served as a unique identifier and was treated as the primary key.

Initial cleaning involved removing extraneous metadata included by Qualtrics—such as survey start and end times, IP addresses, geolocation data, and question display logic—all of which were irrelevant to the analysis. These columns were trimmed to streamline the dataset for subsequent transformation and statistical work.

Column names in the original Qualtrics export were alphanumeric but often ambiguous and misleading. Many variable names did not match the corresponding survey question numbers. Our team manually mapped the exported column names to their corresponding survey questions and responses by referencing adjacent metadata fields and using deductive reasoning. This process allowed us to build an index-based column naming structure, which greatly improved the manageability and interpretability of the dataset.

Before diving into question-specific analysis, we first identified the subset of survey questions relevant to our research objectives. All unrelated or out-of-scope items were removed. This step reduced the employer dataset from 176 columns to 100, and the educator dataset from 171 columns to 102.

Several formatting inconsistencies also needed to be resolved. Some multi-select questions appeared in the form of comma-separated text responses within a single column, while others were exported into multiple binary columns. Additionally, for certain questions, a response option that received zero selections was dropped entirely by Qualtrics. To standardize these issues, we implemented a script to "explode" comma-separated responses into individual binary columns. For dropped columns, we manually reintroduced them as zero-filled dummy variables to preserve the full response structure.

Finally, we filtered out participants who answered less than half of the survey. We also excluded:

- Employers who responded "No" to the question: "Do you work with early career veterinarians (someone who has graduated from a DVM program after May 2021)?"

- Educators who responded “No” to: “Do you teach in any capacity of the dental curriculum at your institution?”

After all preprocessing steps, the final cleaned datasets consisted of 13 employer participants and 30 educator participants.

4 Statistical Methods

Likert scale questions (statistical questions #1,3, 6, and 8)

These questions use a Likert scale with response options: Strongly Agree, Agree, Disagree, and Strongly Disagree across a variety of skills and procedures. Given the ordinal nature of these responses, we will use diverging stacked bar charts and summary tables to visually explore the data. We will then use Mann-Whitney U Tests for formal hypothesis testing differences between groups. Finally, a Ranked-Biserial Correlation (RBC) of effect size to find direction and magnitude of differences between groups.

The Mann-Whitney U Test is a non-parametric test used to assess whether there is a statistically significant difference in the distributions of ordinal or continuous variables between two independent groups. In this case, educators and employers are assumed to be independent, consistent with the survey design.

This test is appropriate for Likert-scale data because:

- It does not assume normality (unlike the t-test),
- It respects the ordinal (ranked) structure of the data,
- It retains statistical power in small samples.

Table 4.1: Mann-Whitney U Rank Summary Table

Group	Sample.Size	Sum.of.Ranks	Mean.Rank
Group 1 (e.g., Educators)	n□	R□	R□ / n□
Group 2 (e.g., Employers)	n□	R□	R□ / n□

In order to calculate the test statistic and p-value for a Mann-Whitney U Test:

- All observations are ranked together
- The sum of the ranks for each group is calculated (R_1 and R_2)
- The test statistic U is then calculated as the minimum of U_1 and U_2 where U_1 and U_2 are the following: $U_1 = R_1 - \frac{n_1(n_1+1)}{2}, U_2 = R_2 - \frac{n_2(n_2+1)}{2}$
- Then the p-value is calculated from the U distribution or normal approximation.

To complement the Mann-Whitney U tests, effect sizes were calculated using the rank-biserial correlation (RBC). While the p-value assesses whether a statistically significant difference exists between groups, the rank-biserial correlation quantifies the magnitude and direction of that difference—adding a layer of practical interpretation. An RBC value near 0 suggests little to no effect, while values near -1 or 1 suggest large differences favoring one group over the other.

Formally, the RBC is calculated directly from the Mann-Whitney U statistic:

$r_{rb} = \frac{2U}{n_1 n_2} - 1$

Where:

- U is the Mann-Whitney U statistic (typically the smaller of U_1 and U_2)

- n_1 and n_2 are the sample sizes of the two independent groups

This formula scales U into the $[-1, 1]$ interval, where:

- $r_{rb} = 0$ implies overlapping ranks (no difference)

- $r_{rb} = 1$ implies every value in group 1 exceeds every value in group 2

- $r_{rb} = -1$ implies every value in group 2 exceeds every value in group 1

U is derived by jointly ranking all observation across groups. The sum or ranks (R_1 and R_2) are then used to calculate:

$U_1 = R_1 - \frac{n_1(n_1+1)}{2}$,
$U_2 = R_2 - \frac{n_2(n_2+1)}{2}$

Then U = min(U_1,U_2) and this value is substituted for r_{rb}.

P-values and an RBC value will be applied to each statistical question listed above in order to formally analyze the data.

ADD MENTION HERE OF EFFECT SIZE CALCULATIONS.

Select all that apply questions (statistical questions #2, 5, and 7)

These questions asked the participants to "select all that apply" as it relates to skills in pre-clinical curriculum, format of dental instruction, and skills for clinical training respectively. We will utilize frequency tables and bar plots to explore the data for these questions and use Fisher's Exact Test as a formal inference procedure for the comparison of the two groups. Dot plots will also be utilized for questions with many options to avoid crowding in bar graphs.

Fisher's Exact Test works with categorical data with independent samples, in this case educators and employers. Based on the survey design from the client, it is again reasonable to assume that these two groups are independent. In this context, Fisher's Exact Test is preferred over Chi-Squared Tests due to the small sample size and therefore not meeting the expected count threshold that is required to proceed with Chi-Squared Tests.

Table 4.2: Fisher's Exact 2×2 Contingency Table

Group	Outcome.Present	Outcome.Absent	Row.Totals
Group 1 (ex: Educators)	a	b	a + b
Group 2 (ex: Employers)	c	d	c + d
Column Totals	a + c	b + d	n = a + b + c + d

Fisher's Exact Test operates under the null hypothesis that there is no association between the two variables. The alternative hypothesis is that there is an association. The p-value for the test is computed using the following formula:

$$p = \frac{(a+b)!\,(c+d)!\,(a+c)!\,(b+d)!}{a!\,b!\,c!\,d!\,n!}$$

P-values are then computed for each skill, format, etc. between the two groups (educators and employers) depending on the statistical question to be analyzed.

Numerical entry questions: (statistical question #4)

This question asks participants to enter a number related to the number of dental procedures that should be completed during training in different areas. We plan to produce box plots and/or histograms to visually examine the data. Depending on the normality or lack thereof of the distributions, we will then conduct either a two-sample t-test or a Mann-Whitney U Test. As mentioned above, the Mann Whitney U test is a non-parametric test that does not depend on the assumption of normality. If the assumption on normality is met, we can consider a two-sample t test for this analysis. However, from examination of the data, normality is not met and we will therefore utilize a Mann-Whitney U Test. The test follows the same method above as statistical questions #2, 5, and 7.

5 Results

5.1 Statistical Analysis

S1 Are there significant differences between educators and practice owners in their belief that new graduates are competent in key dental skills on their first day of practice?

To assess whether there are differences in perceptions of new graduate competence in dentistry, educators and employers were asked a parallel question (Q4). Employers were asked to rate their expectation that early career veterinarians can competently perform 12 specific dental skills on their first day of employment. Educators were asked to rate whether they believe new graduates can perform those same skills competently. Responses were recorded on a 4-point Likert scale from 1 (Strongly Disagree) to 4 (Strongly Agree).

We used the Mann-Whitney U test to compare educator and employer responses for each skill, as this non-parametric test is well-suited for ordinal data from two independent groups. In addition to p-values, we calculated the ranked-biserial correlation (r_rb) to assess effect sizes and provide insight into the magnitude and direction of observed differences. Results are summarized with median ratings for each group, sample sizes, and effect size interpretations (e.g., small, medium, large). Significance stars were added to highlight results below conventional thresholds (e.g., $p < 0.05$), and results were sorted by p-value to aid interpretation.

Table 5.1: Mann-Whitney U-Test Results for Competency Ratings

Skill	Med (Edu)	Med (Emp)	N (Edu / Emp)	P-Value	r_rb	Interpretation
Periodontal probing	4	4	28 / 13	1.000	0.00	Negligible effect
Dental radiography - positioning	3	2	27 / 12	0.369	0.18	Small positive effect
Dental radiography - interpretation	3	3	28 / 12	0.052 .	0.36	Medium positive effect
Scaling	4	4	27 / 13	0.552	-0.10	Negligible effect
Polishing	3	3	28 / 13	0.242	0.21	Small positive effect
Simple extractions - canine	3	3	28 / 13	0.374	0.16	Small positive effect
Simple extractions - feline	3	3	28 / 13	1.000	0.00	Negligible effect
Open extractions - canine (single root)	3	3	28 / 13	0.769	-0.06	Negligible effect
Open extractions - feline (single root)	2	2	27 / 13	0.613	-0.10	Negligible effect
Open extractions - canine (multi-root)	2	3	28 / 13	0.289	-0.20	Small negative effect
Open extractions - feline (multi-root)	1	1	7 / 4	0.778	-0.11	Small negative effect
Fluoride treatment	3	4	2 / 1	1.000	-0.50	Large negative effect

S2 Is there a difference between educators and practice owners in their reports (educators' actual teaching vs. owners' perceptions) of which dental skills were taught in the pre-clinical DVM curriculum for recent graduates?

To evaluate whether differences exist between DVM educators and employers in their understanding of which dental skills are taught in the pre-clinical curriculum, we analyzed responses to parallel survey questions. Educators were asked to indicate which of seven core dentistry skills are taught as part of their pre-clinical courses (Q12), while employers were asked which skills they believe recent graduates were taught prior to clinical training (Q16). Since the two groups answered different question numbers about the same underlying skills, we first aligned the datasets by renaming employer variables to match educator labels. This harmonization allowed for direct comparison of responses skill-by-skill across the two groups.

After cleaning and reshaping the data into long format, we filtered out invalid or missing entries and conducted Fisher's Exact Tests for each skill. This non-parametric test is appropriate for evaluating categorical (yes/no) outcomes in small samples, especially when comparing proportions between two independent groups. Only skills for which both groups provided non-missing responses were included in the final analysis.

Table 5.2: Fisher's Exact Test Results by Skill

Skill	p_value	n_educator	n_employer	prop_educator	prop_employer
Dental radiograph acquisition	0.307	28	13	0.714	0.538
Oral examination and charting	0.399	28	13	0.786	0.923
Simple extraction technique	0.469	28	13	0.750	0.615
Instrument identification and handling	0.732	28	13	0.643	0.538
Dental cleaning (scaling and polishing)	0.742	28	13	0.536	0.615
Local nerve blocks	0.742	28	13	0.607	0.538
Dental radiograph interpretation	1.000	28	13	0.643	0.615

S3 Is there a difference between educators and practice owners in their level of agreement about whether specific dental skills should be taught pre-clinically?

To assess whether educators and employers differ in their beliefs about which dental skills should be included in the pre-clinical DVM curriculum, we analyzed responses to parallel Likert-scale questions: Q13 (educators) and Q17 (employers). Both groups rated the importance of teaching 12 specific dentistry skills using a 4-point Likert scale ranging from 1 (Strongly Disagree) to 4 (Strongly Agree).

Because the data were ordinal and responses came from two independent groups, the Mann-Whitney U test was used to compare educator and employer ratings for each skill individually. This non-parametric test is appropriate for comparing ordinal data between two independent groups, especially when the assumptions for parametric tests may not hold. Ratings were reshaped into long format and analyzed skill-by-skill. The results table presents medians for each group, sample sizes, p-values with significance indicators, effect size estimates (rank-biserial correlation), and an interpretation of the magnitude and direction of group differences.

Table 5.3: Mann-Whitney U-Test Results for Pre-Clinical Dentistry Training

	Median		N		Effect Size	
Skill	**Med (Edu)**	**Med (Emp)**	**N (Edu / Emp)**	**P-Value**	**r_rb**	**Interpretation**
Open extractions - feline (multi-root)	3	4	17 / 13	0.002 **	-0.60	Large negative effect
Fluoride treatment	1	3	3 / 9	0.013 *	-1.00	Large negative effect
Open extractions - canine (multi-root)	4	4	26 / 13	0.079 .	-0.29	Small negative effect
Open extractions - canine (single root)	4	4	26 / 13	0.084 .	-0.29	Small negative effect
Open extractions - feline (single root)	4	4	27 / 13	0.093 .	-0.28	Small negative effect
Simple extractions - feline	4	4	26 / 13	0.136	-0.26	Small negative effect
Dental radiography - interpretation	4	4	28 / 12	0.164	0.13	Small positive effect
Scaling	4	4	28 / 13	0.403	0.14	Small positive effect
Polishing	4	4	28 / 12	0.480	0.12	Small positive effect
Periodontal probing	4	4	28 / 13	0.599	0.04	Negligible effect
Dental radiography - positioning	4	4	27 / 11	0.726	0.07	Negligible effect
Simple extractions - canine	4	4	28 / 13	0.788	-0.05	Negligible effect

S4 Do employers and educators differ in their expectations about how many dental procedures new graduates should complete during clinical training?

To assess whether educators and practice owners differ in their expectations for the number of dental procedures students should complete during clinical training, we compared responses from Question 19 of the educator survey and Question 24 of the employer survey. Respondents indicated the number of procedures they believe should be completed in four clinical settings: primary care, dentistry rotations, shelter medicine, and lab animal medicine.

Before selecting an appropriate statistical test, we assessed normality using the Shapiro-Wilk test, which showed that the data were not normally distributed in any group-setting combination (all p-values < 0.05).

Mann-Whitney U Test: Expected Number of Dental Procedures

Comparison of Educator and Employer Expectations by Clinical Setting

Setting	Med (Edu)	Med (Emp)	N (Edu)	N (Emp)	P-Value	r_rb	Effect Size
Dentistry_Rotation	4.0	6.5	24	10	0.080 .	-0.39	Medium negative effect
Lab_Animal	0.0	0.0	16	9	0.222	0.24	Small positive effect
Shelter_Medicine	2.0	0.0	19	9	0.235	0.27	Small positive effect
Primary_Care	3.0	4.0	27	10	0.287	-0.23	Small negative effect

Given the non-normal distribution and the ordinal or skewed nature of the data, we used the Mann-Whitney U test—a non-parametric alternative to the independent samples t-test—to compare the distributions of responses between educators and employers for each clinical setting. This test is appropriate when comparing two independent groups without assuming normality or equal variances.

S5 Is there difference between the instructional formats in dentistry reported by DVM programs and the formats perceived by employers to have been completed by early career veterinarians?

Both educators and employers were asked a question relating to the format of instruction during the clinical year.

On question 20 of their survey, employers were asked: "What format of clinical instruction in dentistry do you believe that the early career veterinarians (individuals who have graduated from a DVM program after May 2021) hired into your practice/organization/institution completed as part of their DVM training? Select all that apply."

On question 16 of their survey, educators were asked: "What format of instruction in dentistry does your DVM program provide during the clinical year? Select all that apply."

Since the question was of the format "select all that apply," participants were able to select more than one response and percentages will not add to 100%. Results reported below are the percentages of their respective group (educators or employers) that selected the given format of instruction. Percentages were utilized due to the difference in sample size.

Educators and employers were also given the option to enter their own responses to the question. One employer opted to write in "Cadaver." Two educators entered their own responses with one saying "rounds" and the other saying "topic seminars do occur with some rotations during case rounds when dental cases are chosen to present."

Educators and employers differed in their perceptions of clinical dental instruction formats completed by early career veterinarians. While a majority in both groups acknowledged didactic instruction, educators reported higher rates of wet lab (75% vs. 46.2%) and live patient training (96.4% vs. 38.5%) compared to employers. Conversely, employers more frequently identified simulation training (30.8% vs. 14.3%) than educators. These differences suggest varying expectations or awareness between the two groups regarding dental training experiences of recent graduates.

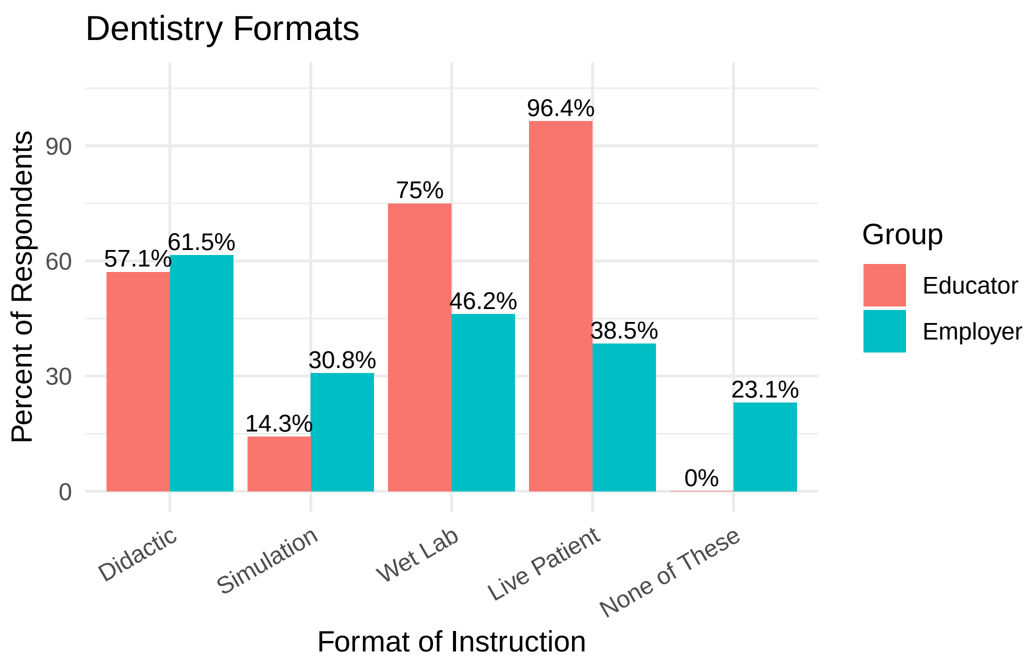

Figure 5.1: Perceived Clinical Instruction Formats in Dentistry Completed by Early Career Veterinarians

Figure 5.1 shows the percentages of each format selected and Table 5.2 shows the p-value associated with performing Fisher's Exact Test on each format.

Didactic instruction was selected by 57.1% of educators and 61.5% of employers, with no statistically significant difference between the groups (p=1). There was also no statistically significant difference in simulation (p=0.237) and wet lab instruction (p=0.0885).

On the other hand, live patient instruction was selected by 96.4% of educators, but only 38.5% of employers. This difference was found to be statistically significant with a p-value of 0.000105.

"None of these" was also found to be statistically significant at the 5% level. No educators selected that none of these formats were believed to have been used, but 23.1% of employers selected that none of the formats were believed to be utilized in the clinical year.

Table 5.4: Perceived Formats of Clinical Dental Instruction in DVM Programs

Format	% of Educators Selected	% of Employers Selected	P-value
Didactic	57.1	61.5	1
Simulation	14.3	30.8	0.237
Wet Lab	75.0	46.2	0.0885
Live Patient	96.4	38.5	0.000105 ***
None of These	0.0	23.1	0.0268 *

S6 Do educators and employers differ in their views on which formats of clinical instruction in dentistry should be required for DVM students as part of their clinical training?

In Question #21 of the employer survey, participants were asked:

• "Which of the following types of clinical instruction in dentistry do you think DVM students should be required to complete as part of a DVM program? Select one response for each of the instructional types listed below."

The analogous question for educators appeared as Question #17 and stated:

• "Which of the following types of instruction do you think DVM students should be required to complete as part of their clinical training? Select one response for each type of instruction listed below."

This question aimed to assess perceptions regarding which types of dental instruction should be required in the veterinary medical curriculum across both survey groups. Table 5.3 presents the Mann–Whitney U-test results, summary statistics, effect sizes, and ranked-biserial correlation outputs.

Table 5.5: Mann-Whitney U-Test Results for Clinical Instruction Formats

	Median		N		Effect Size	
Skill	Med (Edu)	Med (Emp)	N (Edu / Emp)	P-Value	r_rb	Interpretation
Live Patient	4	4	28 / 13	0.004 **	-0.35	Medium negative effect
Simulation	3	4	20 / 12	0.366	0.18	Small positive effect
Didactic	4	4	23 / 13	0.467	0.13	Small positive effect
Wet Lab	4	4	24 / 13	0.658	0.05	Negligible effect

The Mann–Whitney U-test revealed a statistically significant difference ($p = 0.004$) between employers and educators when evaluating the importance of Live Patients. Although both groups reported a median rating of 4, suggesting similar central tendencies, the ranked-biserial correlation ($r_{rb} = -0.35$) indicates a medium negative effect. This suggests that employers generally rated Live Patient instruction as more essential than educators did. The difference in ranking distributions is statistically meaningful.

No statistically significant differences were found between the groups regarding the instructional types Simulation ($p = 0.366$), Didactic ($p = 0.467$), or Wet Lab ($p = 0.658$). The ranked-biserial correlations for Simulation ($r_{rb} = 0.18$) and Didactic ($r_{rb} = 0.13$) were small and positive, indicating a slight tendency for educators to rate these types of instruction as more important than employers. The effect size for Wet Lab ($r_{rb} = 0.05$) was negligible. None of these differences reached statistical significance.

The instructional categories "None of These" and "Other" were excluded from Table 5.3 due to insufficient data for statistical testing.

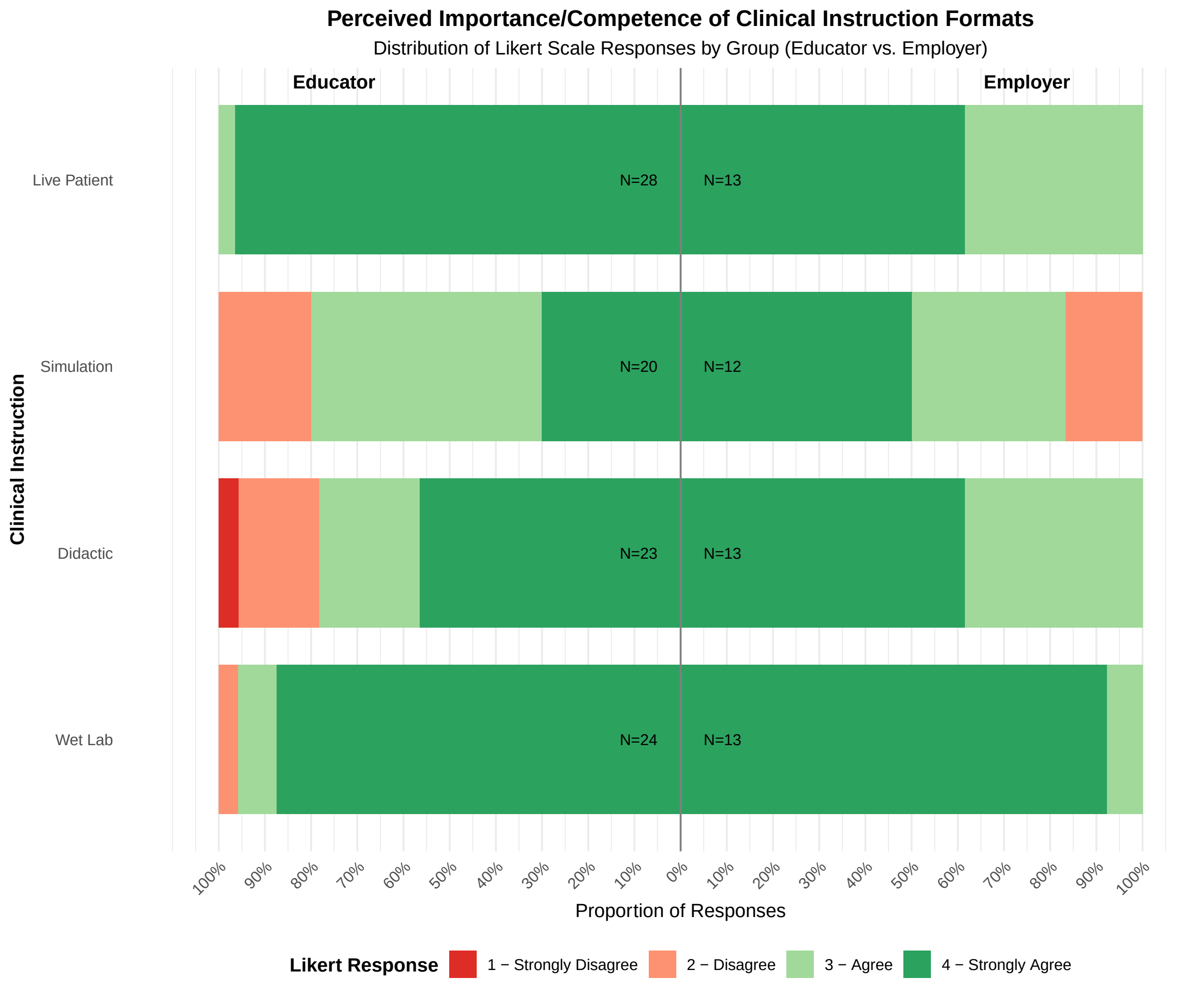

Figure 5.2: Diverging Stacked Bar Chart of Perceived Importance/Competence of Clinical Instruction Formats by Group (Educator vs. Employer).

Figure 5.2 presents the proportional distribution of responses for each group. While a moderate difference was observed between groups regarding the use of live patients, the majority of participants in both groups selected Strongly Agree, and notably, no participants selected Disagree or Strongly Disagree—indicating strong consensus that live patient experience should be required in clinical training.

The response distributions for simulation and wet lab instruction appear nearly identical visually, and statistical analysis confirmed no significant differences between groups for these skills. Interestingly, the distribution for didactic instruction looks somewhat different from those two, despite also showing no statistically significant difference.

This highlights an important point: even when one bar chart appears to show more variability in proportions than another, the overall ranking of responses between groups may still be similar. The Mann-Whitney U-test is designed to detect consistent directional differences in how groups rank responses. If those patterns are weak, inconsistent, or cancel each other out, the test will indicate no significant difference—even when the visualizations suggest otherwise. Differences in sample size and variability can also exaggerate or smooth over apparent contrasts that don't reflect meaningful differences in central tendency.

S7 Is there a difference between the clinical dentistry skills that educators report DVM students are learning during their clinical training and the skills that employers believe recent graduates have completed as part of their DVM program?

Both educators and employers were asked a question related to skills learned and practiced during the clinical year.

On question 25 of their survey, employers were asked: "Which of the following skills do you think that individuals who graduated with a DVM degree after May 2021 completed during the clinical training portion of their DVM program? Select all that apply."

On question 20 of their survey, educators were asked: "Which of the following skills are DVM students at your institution practicing/learning during the clinical training portion of the DVM program? Select all that apply."

Since the question was of the format "select all that apply," participants were able to select more than one response and percentages will not add to 100%. Results reported below are the percentages of their respective group (educators or employers) that selected the given format of instruction. Percentages were utilized due to the difference in sample size.

Educators and employers were also given the option to enter their own responses to the question. No employers entered any text responses. Six educators entered text responses of the following:

- "OvaVet gel application, Crown amputation, Sealant application for UCF, oral tumor biopsy and excision, orthodontia"
- "Nerve blocks, barrier sealant and bonded sealant application, jaw fracture repair, root canals"
- "Often- bonded sealants; Sometimes- root canals, restorations, jaw fracture repair"
- "Oral biopsy, root planning, bonded sealants"
- "extractions only if performed on that patient"
- "Not all students see all types of extractions"

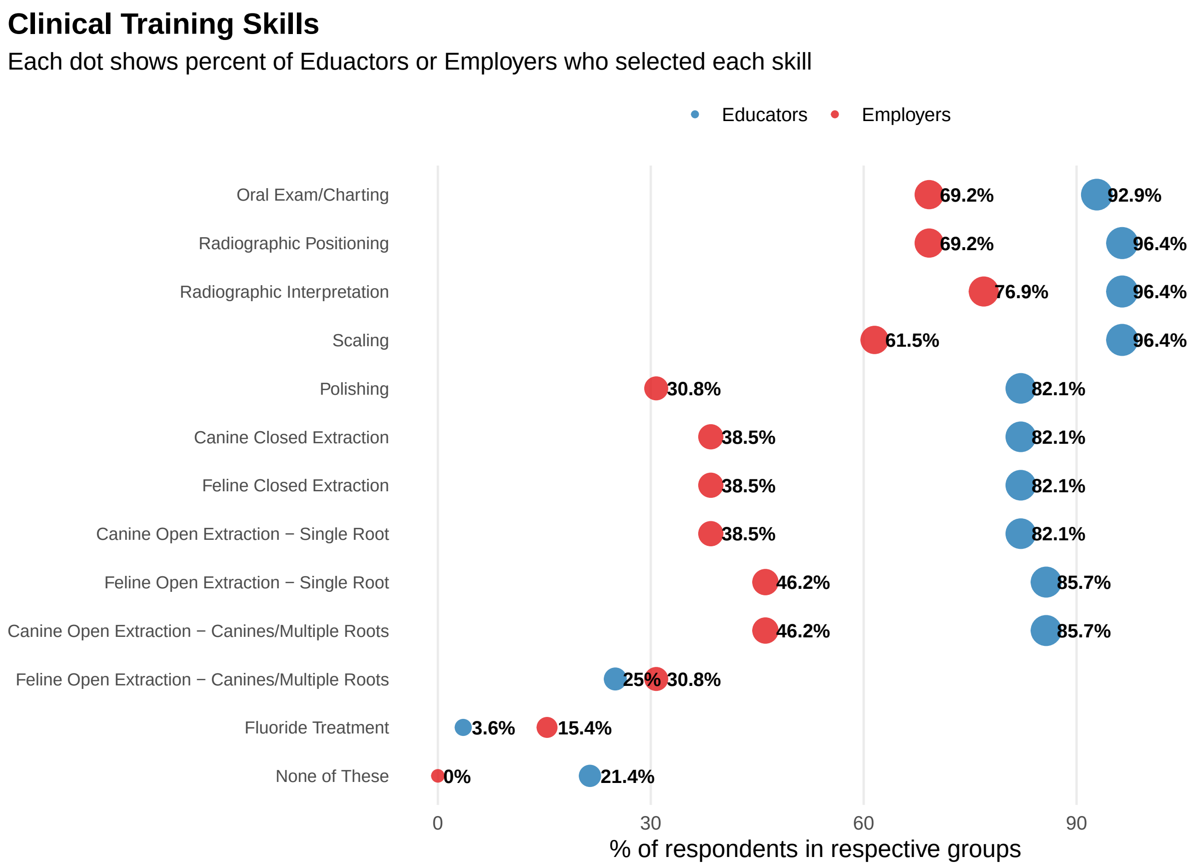

Figure 5.3: Clinical Training Skills Perceived by Educators vs Employers in DVM Programs

Table 5.6: Perceived Skills Learned of Clinical Dental Instruction in DVM Programs

Skill	% of Educators	% of Employers	P-value
Oral Exam/Charting	92.9	69.2	0.0685
Radiographic Positioning	96.4	69.2	0.0284 *
Radiographic Interpretation	96.4	76.9	0.0861
Scaling	96.4	61.5	0.0084 **
Polishing	82.1	30.8	0.00331 **
Canine Closed Extraction	82.1	38.5	0.0102 *
Feline Closed Extraction	82.1	38.5	0.0102 *
Canine Open Extraction - Single Root	82.1	38.5	0.0102 *
Feline Open Extraction - Single Root	85.7	46.2	0.0193 *
Canine Open Extraction - Canines/Multiple Roots	85.7	46.2	0.0193 *
Feline Open Extraction - Canines/Multiple Roots	25.0	30.8	0.719
Fluoride Treatment	3.6	15.4	0.232
None of These	21.4	0.0	0.152

Figure 5.3 shows the percentage of employers and educators who selected the given skills and Table 4 shows the p-values associated with Fisher's Exact Test and their significance.

For eleven of the thirteen skills, educators reported a higher percentage of selection. The two skills for which employers reported higher perceived learning were feline open extraction - canines/multiple roots and fluoride treatment.

At the 5% significance level, the difference in skill selection between employers and educators was found to be significant for seven skills:

- Radiographic Positioning (p=0.0284)
- Scaling (p=0.0084)
- Polishing (p=0.00331)
- Canine Closed Extraction (p=0.0102)
- Feline Closed Extraction (p=0.0102)
- Canine Open Extraction - Single Root (p=0.0102)
- Feline Open Extraction - Single Root (p=0.0193)
- Canine Open Extraction - Canines/Multiple Roots (p=0.0193)

For all skills with statistically significant differences, educators reported higher percentages than employers.

The most significant difference was found in polishing. Educators selected this skill 82.1% of the time, while employers selected this skill only 30.8% of the time. The resulting p-value was 0.00331.

S8 Do educators and employers differ in their opinions about which clinical dentistry skills DVM students should be required to practice or learn during their clinical training?

In question #26 of the employers' version of the survey, participants were asked:

- "Which of the following skills do you think that DVM students should be required to practice/learn as part of the clinical training portion of a DVM program? Select one response for each of the skills listed below."

The corresponding question for educators was survey question #21.

This question aimed to assess opinions on which clinical dentistry skills should be required in DVM student training. Table 5.5 presents the Mann–Whitney U-testresults, summary statistics, effect sizes, and ranked-biserial correlation outputs.

Table 5.7: Mann-Whitney U-Test Results for Clinical Procedures

	Median		N		Effect Size	
Skill	Med (Edu)	Med (Emp)	N (Edu / Emp)	P-Value	r_rb	Interpretation
Feline closed extraction	4	4	26 / 11	0.064 .	0.27	Small positive effect
Canine open extraction - Single root	4	4	26 / 11	0.064 .	0.27	Small positive effect

Feline open extraction - Single root	4	4	26 / 11	0.092 .	0.23	Small positive effect
Canine open extraction - Canines/multiple roots	4	4	26 / 11	0.092 .	0.23	Small positive effect
Radiographic interpretation	4	4	26 / 11	0.261	0.12	Small positive effect
Feline open extraction - Canines/multiple roots	3	4	16 / 9	0.276	0.26	Small positive effect
Canine closed extraction	4	4	26 / 11	0.437	0.11	Small positive effect
Radiographic positioning	4	4	26 / 11	0.510	-0.10	Negligible effect
Scaling	4	4	26 / 11	0.578	-0.09	Negligible effect
Polishing	4	4	26 / 11	0.816	-0.03	Negligible effect
Oral exam/charting	4	4	26 / 11	0.829	0.03	Negligible effect

Nearly every skill reported a median of 4 with exception to the Feline Open Extraction - Canines/Multiple Roots whose central tendency was 3. Note that this exception was the only row that report fewer responses than the others. Small but consistent differences were found for several clinical procedures, with educators tending to rate certain skills as more important than employers.

Specifically, educators rated the following procedures as more important (higher median values) compared to employers:

- Feline closed extraction (effect size: $r = 0.27$)
- Canine open extraction – single root ($r = 0.27$)
- Feline open extraction – single root ($r = 0.23$)
- Canine open extraction – canines/multiple roots ($r = 0.23$)
- Feline open extraction – canines/multiple roots ($r = 0.26$)
- Radiographic interpretation ($r = 0.12$)
- Canine closed extraction ($r = 0.11$)

All of these showed small positive effect sizes, indicating that educators placed slightly more emphasis on these procedures than employers. However, none of these differences reached conventional levels of statistical significance (i.e., $p < 0.05$), though some were marginal ($p \approx 0.06$–0.09). Negligible differences were found for the following skills:

- Radiographic positioning ($r = -0.10$)
- Scaling ($r = -0.09$)
- Polishing ($r = -0.03$)
- Oral examination/charting ($r = 0.03$)

These results suggest that educators and employers generally agreed on the importance of these particular skills, with only very small or no measurable differences between groups.

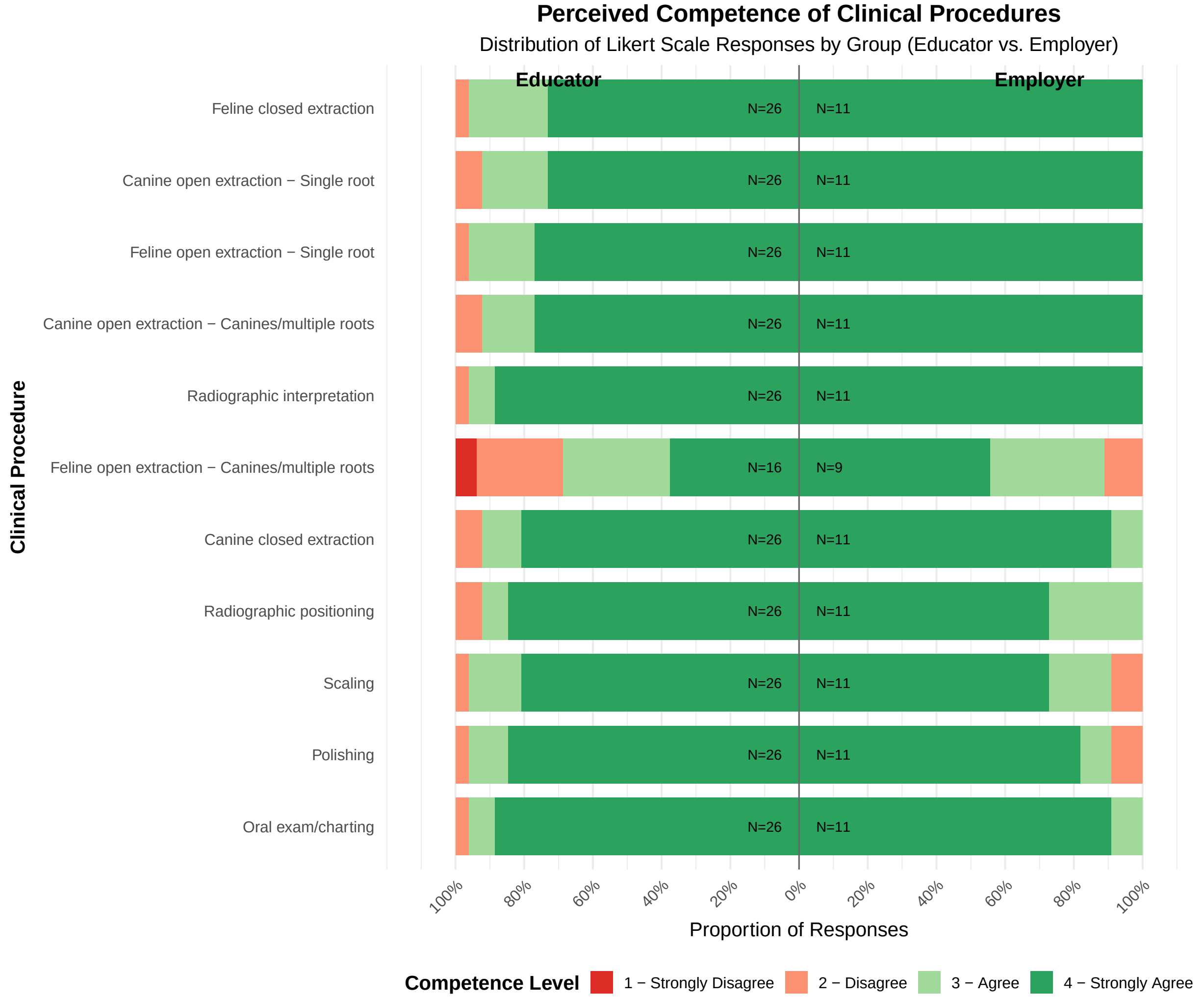

Figure 5.4: Diverging Stacked Bar Chart of Perceived Competence of Clinical Procedures by Group (Educator vs. Employer).

While the effect sizes for several procedures suggest that educators may place slightly more value on certain clinical extraction skills and radiographic interpretation than employers, these differences were not statistically significant. The negligible effect sizes for basic procedures such as polishing and oral exams indicate a strong consensus between the two groups regarding these skills. Figure 5.4 shows that the distribution of responses is broadly similar, and although some differences exist, they do not reach statistical significance to suggest meaningful divergence in perceptions of clinical competencies. Notably, the procedure "Feline Open Extraction – Canines/Multiple Roots" exhibits the greatest variability; however, this may be influenced by the smaller number of responses for that item.

6 Discussion/Conclusion

6.1 Interpretation of Results

6.2 Implications of the Study

6.3 Limitations

6.4 Recommendations

6.5 Summary of Key Findings

6.6 Final Thoughts

Appendix

American Veterinary Medical Association. 2019. "Census of Veterinarians Finds Trends, Shortages, Practice Ownership." 2019. https://www.avma.org/javma-news/2019-07-15/census-veterinarians-finds-trends-shortages-practice-ownership.